# Learning to Contest: Decentralized Robust Fairness in Cooperative MARL via Cross-Attention

Can Savcı
*Independent Researcher*
İzmir, Turkey
savcisavcican@gmail.com

***Abstract*—Fair cooperative multi-agent reinforcement learning (MARL) teams that maximize an egalitarian welfare are exploitable: a single self-interested agent free-rides on the surplus that fair agents forgo to raise the worst-off. A centralized need-based allocator removes the exploit, but only by taking allocation out of the agents' hands; whether *decentralized* policies can be robust was left open. We show that the futility of a decentralized defense is an artifact of *all-or-nothing* contention. Under *graded* contention—where a contested resource still delivers a fraction $1-c$ and wastes $c$—we prove that for any $c < 1$ a worst-off cooperator that contests a free-rider strictly improves on yielding, so decentralized leverage exists (Prop. 1). Realizing it, however, requires solving a coordination problem under uncertainty: the number of free-riders is *unknown and variable*, so any fixed rule is dominated (always-contest wastes $c$ when nobody defects; always-yield collapses when someone does). We introduce CAN, a permutation-equivariant cross-attention policy over agents' *observed behaviour* that infers how many free-riders are present and responds proportionally—turn-taking when none, contesting just enough when some. Trained against an adversarial league (PSRO), CAN keeps best-response exploitability low ($\rho \approx 1.2$–$1.5$, vs. $\rho = N$ unprotected) across the full contention range while wasting essentially nothing when no free-rider is present (efficiency $\approx 1.0$ vs. the $1-c$ of always-contest) and retaining most of that efficiency even *with* a free-rider present ($D \geq 1$ efficiency $0.83$–$0.96$, so its robustness is not bought with waste), thereby approaching the centralized oracle on both axes *without* a central allocator. On the same game the fair-MARL learners fail on complementary axes—GGF/FEN learn to yield and are exploitable ($\rho \approx N$), SOTO learns to all-contest and is wasteful (waste $c$)—while CAN is the only method that is both robust and efficient. On two further games we find a clear scope rather than blanket generality: CAN stays efficient ($D=0$ efficiency $0.94$–$1.0$) and Pareto-dominates the welfare-fair learners throughout, but its robustness holds only in proportion to the contest leverage a game provides—strong on a multi-server game, partial when leverage weakens (high-value, high-contention steps), and absent under a winner-take-all rule where Prop. 1 fails. We further report where the approach is fragile—weak leverage and zero-shot transfer to much larger teams both degrade it at high contention—giving an honest account of how far decentralized robust fairness currently reaches.**

***Keywords*—multi-agent reinforcement learning, fairness, exploitability, cross-attention, decentralized execution, league training**

## I. Introduction

Cooperative multi-agent reinforcement learning (MARL) increasingly optimizes not only efficiency but *fairness*: methods such as FEN [1] and SOTO [2] maximize a fairness-inducing welfare (typically the Generalized Gini Welfare [3]) over agents' returns so that no agent is starved. These methods assume every agent cooperates in being fair. When one does not—one stakeholder upgrades a controller to minimize its own delay, one client maximizes its own throughput—the fair team is exploitable: because team-oriented agents *sacrifice* their own utility to raise the worst-off, a self-interested agent free-rides on that sacrifice [4]. In a strictly rivalrous (all-or-nothing) resource—where a contested unit is either won by one agent or wasted—this exploit is hard to defend against *at the policy level*: a cooperator that contests a free-rider merely causes a collision, gaining nothing, so a welfare-fair team is indifferent between yielding and contesting; a recent study formalizes this futility and the centralized need-based remedy that sidesteps it [5]. The robust remedy is then to take allocation out of the agents' hands entirely, via a *centralized* need-based mechanism. This paper takes up the harder question that centralization sidesteps: *can decentralized policies—acting only on local, observed information—retain fairness under a self-interested agent?*

We answer affirmatively, with caveats, in three steps.

**(1) Leverage exists once contention is graded.** The policy-level futility above hinges on *all-or-nothing* contention: a contested resource is either won by one agent or wasted, so a cooperator that contests a free-rider gains nothing. We introduce a *graded-contention* game in which $m \geq 2$ claimers split a fraction $1-c$ of the resource (waste $c$). We prove (Prop. 1) that for any $c < 1$ a worst-off cooperator that contests a lone free-rider receives $(1-c)/2 > 0$ instead of 0, so contesting *strictly dominates* yielding: decentralized leverage reappears. All-or-nothing is the limiting case $c=1$.

**(2) The residual difficulty is coordination under uncertainty.** Leverage is not a policy. The number of free-riders $D$ is unknown and varies episode to episode (including $D=0$). A fixed rule cannot win: always-contest pays the waste $c$ even when nobody defects, and always-yield collapses the moment someone does. A robust cooperator must *infer* $D$ from observed behaviour and respond *proportionally*—do nothing (turn-take) when $D=0$, contest just enough when $D \geq 1$.

**(3) A cross-attention policy, trained against a league, realizes it.** We introduce **CAN** (Cross-Attention Networks), a shared, permutation-equivariant policy in which each agent attends over the *observed behaviour* of all agents (utilities, claim-rates, who is worst-off) to estimate contention and act. What is load-bearing is *behaviour-conditioning with adaptive aggregation*—the response must depend on *how many* others are grabbing, not on fixed identities—and among permutation-respecting aggregators cross-attention is the most stable instance (Sec. V-G). We train CAN against an adversarial league (PSRO [6]) and audit it with a fresh best-response defector. CAN keeps exploitability low ($\rho$≈1.2–1.5) across all contention levels while wasting essentially nothing when no free-rider is present (efficiency ≈1.0), approaching the centralized oracle without any central allocator.

We are deliberate about limits. We report a negative-leaning result—single adversary co-training is unreliable at low contention—and a genuine fragility: zero-shot transfer of an $N$=6 policy to teams of 12 and 24 holds at low contention but degrades at high contention. The contribution is a controlled, honest map of how far decentralized robust fairness reaches, not a claim that the decentralization gap is closed.

## II. Background and Related Work

**Welfare-based fair MARL.** Given per-agent returns $\mathbf{u} = (u_1, \dots, u_N)$, fair-MARL maximizes a Schur-concave welfare $W(\mathbf{u})$ that is increasing (efficiency) and favours equity, e.g. the Generalized Gini Welfare $W(\mathbf{u}) = \sum_k w_k u_k^{\uparrow}$ with decreasing weights $w_k$ so the worst-off agent dominates [3]. FEN [1] and SOTO [2] are representative learners. These objectives create the appropriable surplus we must defend.

**Exploitability and the decentralization gap.** The vulnerability is a fairness-specific instance of free-riding in sequential social dilemmas [7]; that welfare-maximizing cooperation is gameable in general was noted by Ivanov et al. [4], who restore cooperation through an incentive-compatible mediator. Under strictly rivalrous (all-or-nothing) contention, policy-level reciprocity has no efficient lever, and a centralized need-based allocator—which decides who receives each resource—is an upper bound on fairness that any decentralized method should aim to match [5]. We take that target seriously and pursue the decentralized side it leaves open.

**Reciprocity and conditional cooperation.** Inequity aversion [8] and learned reciprocity [9] make cooperation conditional on others' behaviour in social dilemmas. Our cooperators are likewise behaviour-conditioned, but the obstacle here is sharper: under all-or-nothing contention reciprocity has no efficient lever at all; we show graded contention restores one and that a learned attention policy can exploit it.

**Attention in MARL.** Attention over agents is well established for value/critic mixing and communication, e.g. actor-attention critics [10] and the centralized critics of MADDPG [11]. We use a single-head self-attention block [12] *in the decentralized policy* over observed behaviour tokens; its role is contention inference, and its permutation-equivariance gives a single $N$-agnostic policy that can be evaluated at unseen team sizes.

**League / population training.** Training against a growing population of past best responses—fictitious play [13] and PSRO [6]—yields policies robust to a history of adversaries rather than to a single co-evolving one. We use it to stabilize the cooperator against the worst free-rider it has seen, which we find is what makes robustness consistent across contention levels.

## III. The Graded-Contention Game

**Game.** $N$ agents act over $T$ steps. Each step every agent chooses Claim (1, the self-interested act) or Yield (0). Let $\mathbf{u} \geq 0$ accumulate the resource each agent has collected and let $m$ be the number of claimers this step. The unit resource is allocated by the *graded-contention* rule

$$g(m) = \begin{cases} 1 & m = 1, \\ (1-c)\,/\,m & m \geq 2, \\ \text{routed to the worst-off agent} & m = 0, \end{cases} \qquad (1)$$

where each claimer receives $g(m)$ and $c \in [0, 1]$ is the contention *waste*. A sole claimer wins the whole unit; $m \geq 2$ claimers split a discounted $1-c$; if nobody claims, the resource flows (losslessly) to the worst-off agent. The parameter $c$ interpolates from lossless sharing ($c$=0) to the all-or-nothing model ($c$=1, where contesting destroys the resource).

**Threat model.** Each episode, each agent is independently a *defector* with some probability, so the number of free-riders $D$ is variable and unknown (including $D$=0); defectors always claim. The remaining *cooperators* share one policy and seek to keep the allocation fair and efficient. We audit a trained cooperator policy by *freezing* it and training a fresh best-response defector to maximize its own utility—an exploitability test stronger than a fixed always-claim adversary.

**Bounded free-ride metric.** We measure exploitation by the free-ride factor of the defector *group*: its share of the delivered total, normalized by its fair share $n_{\text{def}}/N$,

$$\rho \;=\; \frac{N \sum_{i \in \mathcal{D}} u_i}{n_{\text{def}} \sum_j u_j} \;\in\; [0,\, N/n_{\text{def}}], \qquad (2)$$

where $\mathcal{D}$ is the defector set. $\rho$=1 is exactly fair; $\rho$=$N$ means a lone defector took everything. Eq. (2) is *bounded* by construction, unlike the naive ratio of defector utility to mean-cooperator utility, which diverges when cooperators are starved as $c \to 1$; the delivered total is always positive, so $\rho$ stays in range and is comparable across $c$. We also report Jain's index [14] and efficiency (delivered/$T$).

**Leverage.** Under all-or-nothing contention ($c$=1) a cooperator cannot deny a free-rider except by wasteful levelling-down—contesting collides and destroys the unit—so a welfare-fair team is indifferent between yielding and contesting and has no efficient defense. Graded contention breaks that premise.

*Proposition 1 (Decentralized leverage under graded contention):* Fix $c < 1$ and a step at which a single free-rider claims. Let a worst-off cooperator choose Claim rather than Yield. Then the cooperator's received resource rises from 0 to $(1-c)/2 > 0$, while the free-rider's falls from 1 to $(1-c)/2$. Hence contesting *strictly dominates* yielding for the worst-off cooperator, and strictly reduces the free-rider's appropriated surplus.

*Proof.* If the worst-off cooperator yields, the free-rider is the sole claimer ($m$=1), receives $g(1)$=1, and the cooperator receives 0. If the cooperator contests, $m$=2 and each receives $g(2)$=$(1-c)/2$. Since $c < 1$, $(1-c)/2 > 0$, so the cooperator strictly gains and the free-rider strictly loses. □

The dominant action is also *objective-aligned*: lifting the worst-off cooperator from 0 to $(1-c)/2$ raises the cooperator mean and lowers their spread, so it strictly increases the training welfare $W_{\text{coop}}$=mean−std (Sec. IV). Contesting is thus rewarded both at the level of the individual worst-off agent and of the team objective.

Proposition 1 establishes that the lever *exists* for every $c < 1$; it does not say how to pull it. The cooperator must contest *only* when a free-rider is present (else it pays the waste $c$ for nothing) and only *enough* cooperators must contest (over-contesting wastes; under-contesting leaves surplus). Because $D$ is unknown and variable, this is an inference-and-coordination problem, which motivates a behaviour-conditioned learned policy.

## IV. Method: Cross-Attention Networks (CAN)

**Observed-behaviour tokens.** At step $t$ each agent $i$ is represented by a token of six features computable from public state—no access to others' intentions or rewards:

$$x_i = \left[\tfrac{u_i}{T},\ \tfrac{u_i-\bar{u}}{T},\ \tfrac{u_i-u_{\min}}{T},\ \mathbf{1}[u_i{=}u_{\min}],\ \tfrac{cc_i}{t},\ \tfrac{t}{T}\right], \quad (3)$$

where $\bar{u}, u_{\min}$ are the team mean/min utilities and $cc_i/t$ is agent $i$'s running claim-rate. The claim-rate channel is what lets a cooperator estimate *how many* others are grabbing.

**Architecture.** CAN is a single-head self-attention block over the $N$ tokens: with $Q, K, V$ linear projections of $\{x_i\}$, $\text{ctx} = \text{softmax}(QK^\top/\sqrt{d})V$; each agent concatenates its token with its context, passes a tanh layer, and outputs Claim/Yield logits (Fig. 1a). The block is shared by all cooperators and *permutation-equivariant*, so (i) the policy is decentralized—agent $i$ acts on the public tokens, no central allocator decides the allocation—and (ii) it is $N$-agnostic, enabling zero-shot evaluation at unseen team sizes.

**Cooperator objective.** Cooperators maximize a per-step welfare reward-to-go with welfare

$$W_{\text{coop}} = \text{mean}_{i\in\mathcal{C}}(u_i) - \text{std}_{i\in\mathcal{C}}(u_i), \quad (4)$$

over cooperators $\mathcal{C}$: the mean term rewards reclaiming utility from a free-rider (contest when $D \geq 1$), the std term rewards turn-taking (share evenly), and doing nothing when $D$=0 avoids the waste $c$. Training is REINFORCE with an annealed entropy bonus; the defector(s) during training draw $D \sim \text{Unif}\{0, \ldots, d_{\max}\}$ per episode so the policy sees a variable, unknown adversary count.

**Training schemes.** We compare three ways to provide the adversary, holding the cooperator architecture fixed: *(i) single co-training*—cooperators and one co-evolving defector policy trained together; *(ii) population co-training*—a population of $K$=4 co-evolving defectors, one assigned per episode; *(iii) league (PSRO)* [6]—alternate training the cooperators against the frozen pool of all past best-response defectors and adding a fresh best response to the pool (Fig. 1b). League training targets the multi-equilibrium fragility of single co-training: the cooperator becomes robust to the whole adversary history rather than to one moving target.

**Baselines and oracle.** Two fixed scripts bracket the problem: *all-contest* (everyone always claims; $\rho\approx1$ but efficiency $1-c$) and *yield* (cooperators always yield; efficiency 1 but $\rho$=$N$). The *centralized oracle* allocates by need (worst-off first), achieving $\rho$=1 at efficiency 1—the unbeatable upper bound a decentralized policy aims to approach.

## V. Experiments

**Setup.** Unless noted, $N$=6, $T$=100, $d_{\max}$=2, $c \in \{0.3, 0.5, 0.7, 0.9\}$, 3 seeds; all policies are shared single-head cross-attention nets (hidden 64) trained with REINFORCE/Adam (lr $3\times10^{-3}$, $B$=512 parallel episodes). Every reported policy is audited by a freshly trained best-response defector (Eq. (2)). Implementation details are in the appendix.

### A. Head-to-head with fair-MARL learners

We first place CAN against the welfare-fair learners that motivate the problem, *on the same graded-contention game and under the same bounded audit*: a shared-policy GGF maximizer, a faithful SOTO (Self-/Team-Oriented sub-nets, annealing $\beta$:1→0) [2], and a FEN-style learner whose agents maximize the fair-efficient reward $\bar{u}/(c_0{+}|u_i{-}\bar{u}|)$ [1]. All three see the same six behaviour features and are trained *cooperatively* (no defector at train time, as in their originals); we then freeze each team and train a best-response defector. This comparison is therefore *fair-as-published vs. robust-by-design*: it isolates whether the fair *objective alone* confers robustness (it does not), not whether these objectives could be hardened by adversarial training. The training axis itself—cooperative vs. league—is isolated separately in Sec. V-F on CAN's own architecture.

The result is a clean two-axis separation (Fig. 2, Table I). Every fair learner is fair at $D$=0 (Jain ≈1.0), but each collapses onto one of the two scripted extremes and fails on one axis. **GGF** learns to *yield*: efficient ($D$=0 eff. 1.00) but maximally exploitable ($\rho$=6=$N$ at every $c$—a lone defector takes everything). **SOTO** learns to *all-contest*: robust ($\rho$=1.0) but wasteful—its $D$=0 efficiency is exactly $1-c$ (down to 0.10 at $c$=0.9), because its "be selfish first, then anneal to fair" schedule settles in a fair-by-symmetry but resource-destroying equilibrium. **FEN** is unstable, landing in *either* bad corner depending on seed and $c$ (at $c$=0.5 most seeds learn yield, $\rho\approx$5.9, a minority all-contest, $\rho$=1.0, giving a high-variance mean); its fair-efficient reward does not reliably resolve the tradeoff. **CAN** is the only method in the good corner: efficient (0.98) *and* robust ($\rho$=1.2–1.5), clustered beside the centralized oracle. The remaining experiments isolate the two properties that put it there.

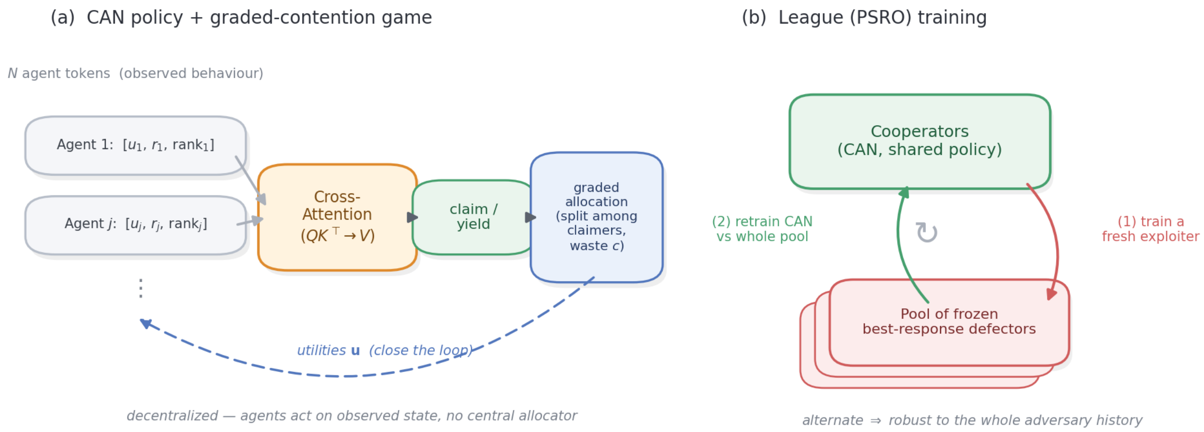


Fig. 1: Overview. (a) The CAN policy: each agent is a token of observed behaviour (utilities, claim-rate, who is worst-off); a shared cross-attention block infers contention and outputs claim/yield; the graded allocator returns utilities, closing a fully *decentralized* loop. (b) League (PSRO) training: alternate retraining the cooperators against the frozen pool of past best-response defectors and adding a fresh exploiter, yielding robustness to the whole adversary history.

TABLE I: Head-to-head on the graded-contention game ($N$=6, 5 seeds). Each fair learner fails on one axis; CAN attains both. Ranges span $c \in \{0.3, \ldots, 0.9\}$.

| Method | $D$=0 eff. | best-resp. $\rho$ | outcome |
|---|---|---|---|
| GGF (welfare) | 1.00 | 6.0 | efficient, exploitable |
| FEN | 0.68–0.98 | 3.0–6.0 | unstable, mostly exploitable |
| SOTO | $1-c$ | 1.0 | robust, wasteful |
| **CAN (league)** | **0.98** | **1.2–1.5** | **both** |
| centralized oracle | 1.00 | 1.0 | (upper bound) |

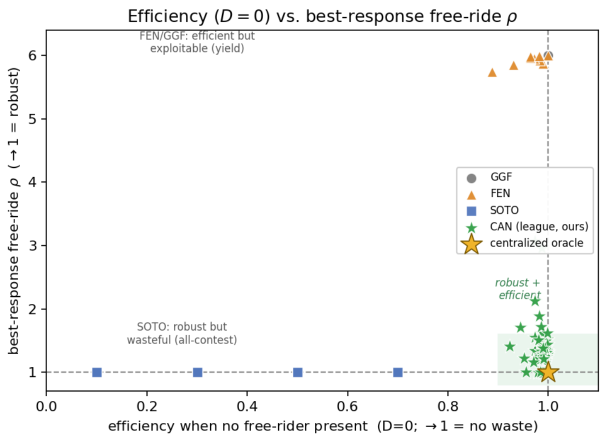


Fig. 2: Efficiency vs. exploitability (each marker is one seed$\times c$). The welfare-fair learners collapse onto the scripted extremes—GGF/yield (efficient, $\rho$=$N$) and SOTO/all-contest (robust, eff. $1-c$)—and FEN scatters between them. Only CAN occupies the good corner (shaded), beside the centralized oracle.

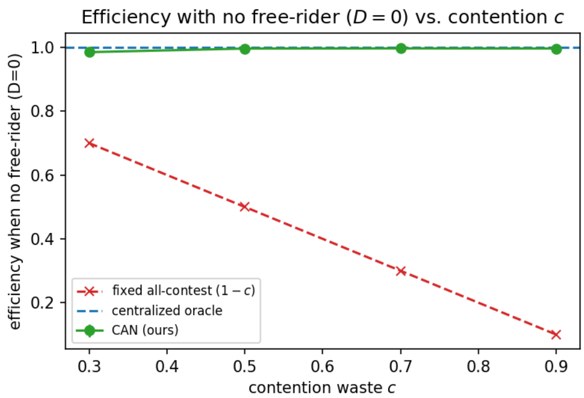


Fig. 3: Efficiency with no free-rider present ($D$=0). CAN turn-takes and wastes essentially nothing (≈1.0) at all $c$, while a fixed all-contest policy pays the full waste $1-c$. Mean over 5 seeds.

## B. *Efficiency when no free-rider is present*

A robust cooperator must not pay for a defense it does not need. Fig. 3 shows the $D$=0 efficiency of CAN against the fixed all-contest script. CAN delivers 0.99–1.00 of the resource at *every* contention level, whereas all-contest delivers only $1-c$ (down to 0.10 at $c$=0.9). Because CAN turn-takes when it detects no free-rider, it incurs essentially no contention waste—matching the centralized oracle on efficiency. This is the half of the problem fixed rules cannot get right.

## C. *Efficiency* under *a defector: is robustness free?*

$D$=0 is the easy case. The axis where a decentralized policy could lose to the centralized oracle—which delivers $\rho$=1 and efficiency 1 *simultaneously*—is efficiency *with* a free-rider present, since contesting forces $m \geq 2$ and delivers only $1-c$ on contested steps; in principle a policy could buy a low $\rho$ at $c$=0.9 by contesting so hard that it wastes as much as the all-contest equilibrium it is meant to beat. It does not. Under the best-response defector, league-trained CAN keeps $D\geq1$ efficiency at 0.88, 0.88, 0.83, 0.96 for $c$=0.3, 0.5, 0.7, 0.9 (pooled over 10 seed-runs)—at the hardest setting it delivers 0.96 versus the all-contest equilibrium's $1-c$=0.10. CAN contests *selectively* (just enough, and mostly via turn-taking rather than collisions), so its robustness is essentially free: it sits near the oracle on *both* axes even with a defector present, where SOTO reaches $\rho$=1 only by destroying 90% of the resource. A residual gap to the oracle remains (efficiency $<$ 1), as expected for a decentralized policy, but it is small, not the structural $1-c$ collapse.

TABLE II: Best-response free-ride $\rho$ vs. contention $c$ ($N$=6, mean with 95% bootstrap CI; pooled over independent gen-6 runs—league 15, vanilla 10 seed-runs per $c$. Lower is better, 1 = fair, oracle = 1.0).

| $c$ | 0.3 | 0.5 | 0.7 | 0.9 |
|---|---|---|---|---|
| vanilla co-train | 1.90 | 2.03 | 2.06 | 2.24 |
| 95% CI | [1.83,1.96] | [1.98,2.09] | [1.97,2.15] | [2.07,2.42] |
| CAN (league) | 1.33 | 1.36 | 1.51 | **1.20** |
| 95% CI | [1.26,1.42] | [1.22,1.53] | [1.25,1.85] | [1.10,1.32] |

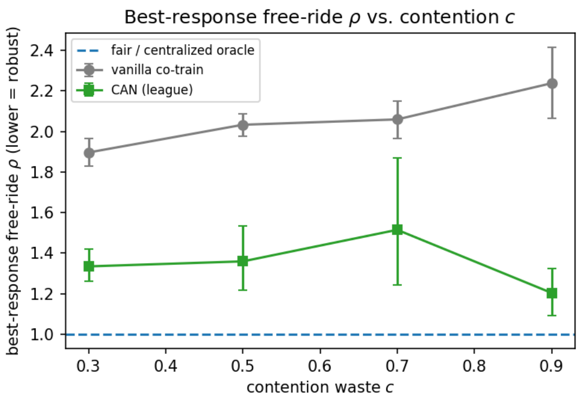


Fig. 4: Robustness vs. contention. League-trained CAN keeps $\rho$ low (1.20–1.51) and is most robust at the highest contention, while the vanilla co-trained policy is roughly twice as exploitable and drifts upward. The fair / centralized-oracle line is $\rho$=1. Mean with 95% bootstrap CI, pooled over independent gen-6 runs (15 league / 10 vanilla seed-runs).

### D. Robustness across the contention range

Fig. 4 and Table II report best-response exploitability $\rho$ vs. $c$ for CAN trained with vanilla co-training (against a fixed always-claim adversary) versus league (PSRO) training (against a growing pool of learned exploiters; to average over run-to-run variance we pool three independent gen-6 league runs, 15 seed-runs per $c$). League training keeps $\rho$ low across the entire range (1.20–1.51), far below the vanilla co-trained policy (1.90–2.24) and well below the unprotected $\rho=N=6$. It is in fact *most* robust at the highest contention ($\rho$=1.20 at $c$=0.9, the lowest point)—consistent with leverage: at higher waste a single contest caps the free-rider's per-step take at $(1-c)/2$, so targeted contention denies it more effectively (Prop. 1). Vanilla co-training shows the opposite drift, rising to 2.24: training against a static adversary does not teach *when* to contest. Training against a *learned* exploiter is what converts the existence of leverage (Prop. 1) into robustness.

### E. Larger $d_{\max}$ and defector coalitions

The main results cap the per-episode defector count at $d_{\max}$=2, an easy "0/1/2 grabbers" inference. We stress this by training CAN with $d_{\max}$=4 (up to four of six defecting) and auditing against a best-response *coalition* of $D$ defectors sharing one policy. CAN remains robust at all $c$ and $D$ (Table III): the worst case is a *lone* free-rider ($\rho$=1.13–1.60, comparable to the $d_{\max}$=2 results), and larger coalitions are *easier*—$\rho\to1.0$ for $D\geq2$—because with more agents defecting there is less cooperative surplus to appropriate. The harder inference and coordinated free-riders thus do not break the policy.

TABLE III: Larger-$d_{\max}$ stress: best-response $\rho$ for CAN trained at $d_{\max}$=4, audited against a $D$-defector coalition ($N$=6, 5 seeds).

| $c$ | $D$=1 | $D$=2 | $D$=3 |
|---|---|---|---|
| 0.3 | 1.22 | 1.11 | 1.07 |
| 0.5 | 1.13 | 1.00 | 1.00 |
| 0.7 | 1.51 | 1.00 | 1.00 |
| 0.9 | 1.60 | 1.05 | 1.00 |

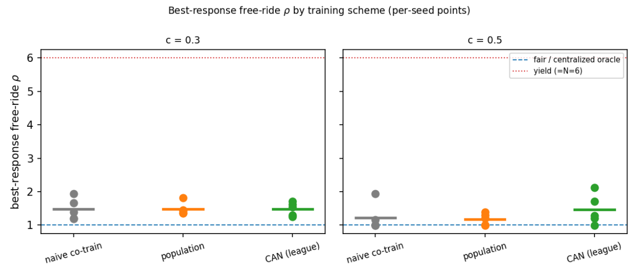


Fig. 5: Per-seed best-response $\rho$ for three adversarial training schemes (5 seeds). The schemes are indistinguishable (all $\rho\approx$1.2–1.5); what matters is training against a *learned* exploiter rather than a static one. The dotted line marks the all-yield ceiling ($\rho=N$).

### F. Choice of adversarial training scheme

Does the result depend on the *league* specifically? Fig. 5 compares three adversarial schemes—single co-training (one co-evolving defector), population co-training ($K$=4 defectors), and the league—at the two lower contention levels, per seed. The schemes are statistically indistinguishable: at $c$=0.3 all three average $\rho\approx$1.48, and at $c$=0.5 they span 1.18–1.46 with overlapping CIs. What carries the result is *that* the cooperator is trained against a *learned*, adapting exploiter at all: every adversarial scheme lands near $\rho$=1.2–1.5, versus 1.8–2.2 for the static-adversary vanilla baseline (Table II). We adopt the league for its robustness to the whole adversary history, but the headline robustness is not an artifact of that choice.

### G. What is load-bearing? Architecture ablation

Is the win the attention, or just behaviour-conditioning with *some* adaptive aggregation? We test this by replacing the cross-attention block with three alternative aggregators of the *same* behaviour tokens, under *identical* league training and the same best-response audit: a *mean-pool* policy (each agent conditions on its token and the mean of all tokens), a *deep-sets* policy (mean-pool of a learned per-agent embedding), and a bidirectional *GRU* over the agent tokens. Table IV reports best-response $\rho$ (mean over $c$, 5 seeds).

The clear conclusion is that *aggregation matters but pooling is not enough*: the mean-pool and deep-sets policies are markedly more exploitable than CAN ($\rho$=1.78, 1.75 vs. 1.37), so the natural objection "a permutation-equivariant pool should already match attention" does not hold. The gap to the strongest alternative is narrower and we do not overstate it: the bidirectional GRU is competitive on the mean (1.56 vs. 1.37) and even better at low $c$, so we credit *behaviour-conditioning plus adaptive aggregation*, not attention per se. Where cross-attention earns its place is *stability*: at the hardest contention ($c$=0.9) CAN holds $\rho$=1.16±0.20 while the GRU is erratic (1.99±1.50), so attention is the most stable instance of the family rather than

TABLE IV: Architecture ablation under identical league training: best-response $\rho$ (mean over $c$, 5 seeds; lower is better). Pooling policies (†, run at a lighter budget) are more exploitable than CAN *even when CAN is held to that same lighter budget* ($\rho$=1.57); at full budget CAN improves to 1.37 and also beats the bi-GRU, chiefly by staying stable at the hardest contention. CAN's $\rho$ in this ablation (1.37) is one of the three independent gen-6 league runs pooled in Table II (pooled mean 1.35); all audit with a fresh best-response defector, and the spread reflects run-to-run (GPU) variance.

| Policy (aggregator) | best-resp. $\rho$ | $\rho$ at $c$=0.9 |
|---|---|---|
| mean-pool† | 1.78 | 2.05 |
| deep-sets† | 1.75 | 2.06 |
| bidirectional GRU | 1.56 | 1.99±1.50 |
| **cross-attention (CAN)** | **1.37** | **1.16±0.20** |

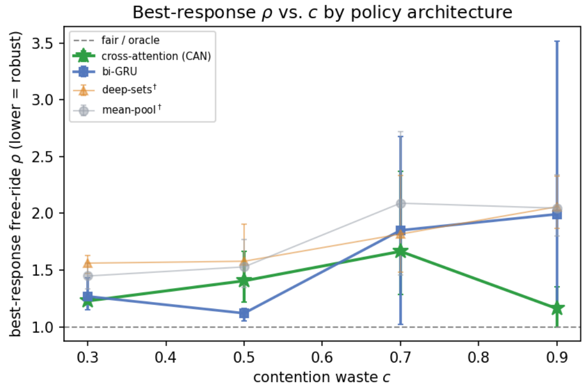


Fig. 6: Architecture ablation: best-response $\rho$ vs. $c$ by aggregator (5 seeds, 95% bootstrap CI). Cross-attention is the only policy that stays low *and* becomes most robust at the hardest contention ($c$=0.9); the bi-GRU is competitive at low $c$ but erratic at high $c$, and the pooling baselines (†, lighter budget) are uniformly worse.

categorically superior. These comparisons are at the deployed training budget; we found that under a *lighter* budget the CAN–GRU ranking reverses, underscoring that architecture ablations must use the budget at which the method is actually trained.

### H. Zero-shot transfer across team size (a limitation)

Because CAN is permutation-equivariant it can be evaluated at team sizes it never trained on. Fig. 7 applies the $N$=6 policy to teams of 12 and 24. Transfer holds well at low contention—at $c$=0.3 a 4× larger team is still close to fair ($\rho$=1.85 at $N$=24)—but degrades at high contention, reaching $\rho$=8.8 at $N$=24, $c$=0.9. The bounded metric keeps these readable (all curves stay below the all-yield ceiling $\rho$=$N$), and the ordering is clean: difficulty grows with both team size and waste. We report this as a genuine fragility—the contention-inference learned at $N$=6 does not fully generalize to crowds at high waste—rather than smoothing it over. The obvious remedy does not work: training on a size curriculum ($N \in \{4, 6, 8, 12\}$, evaluating at the held-out $N$=24) helps only at $c$=0.3 and *worsens* mid-range transfer (it dilutes the per-size budget), leaving the high-contention gap unchanged ($\rho$=4.5 at $c$=0.9, vs. 4.5 for $N$=6-only). Closing it is open.

### I. Generality across environments

The experiments so far use the single-resource graded game. To test whether CAN's behaviour-conditioned contention inference is specific to that game, we replicate the

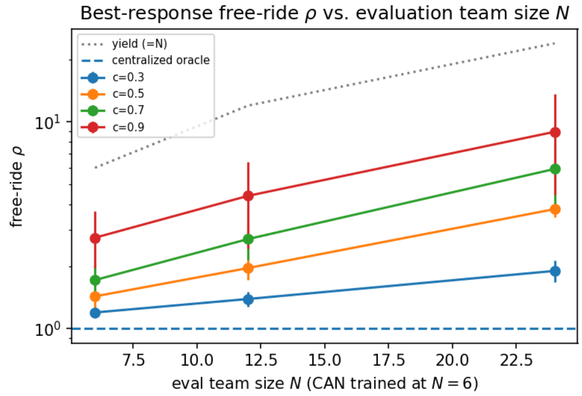


Fig. 7: Zero-shot transfer of the $N$=6 policy to larger teams (log scale). Robust at low contention; degrades with team size at high contention. All curves remain below the all-yield ceiling $\rho$=$N$; the oracle line is $\rho$=1.

head-to-head on two further leverage-preserving environments that pose *different* inference problems (Fig. 8, Table V). *(i) Congestion*: $M$=3 parallel servers; each agent claims one or yields; equal-split per server; empty servers route to the neediest. The cooperators must infer *which* server each free-rider targets and contest just that one. *(ii) Stakes*: a single resource whose value $v_t$ is random (lumpy: usually small, occasionally a jackpot, $\mathbb{E}[v]$=1). The cooperators must contest *selectively*, on the high-stakes steps where a free-rider's surplus concentrates. Both keep equal-split contention, so Prop. 1 leverage holds; CAN and the three welfare-fair learners are trained and audited exactly as before.

**The efficiency axis is what carries over.** On exploitability alone the games differ in difficulty (the congestion exploit is structurally capped at $\rho$=$N/M$=2; stakes restores the $\rho$=$N$ ceiling). The one property that is consistent is efficiency: **CAN keeps** $D$=0 **efficiency** 0.94–1.00 **in all three games**, so it never buys robustness by wasting the resource—unlike the only baseline that attains low $\rho$, SOTO, whose all-contesting collapses efficiency to 0.40–0.47 (and to 0.10 at stakes $c$=0.9). GGF and FEN are Pareto-dominated by CAN in every game. Robustness, however, does *not* carry over uniformly: CAN is both efficient and robust on base and congestion, but on stakes it is only *partially* robust—robust up to moderate contention and exploited at $c$=0.9—so we claim a clear scope, not blanket generality.

**An honest gradient: leverage strength governs success.** CAN is not uniformly robust; its exploitability tracks how much leverage Prop. 1 actually delivers. With *strong* leverage (base: a contest caps the free-rider at $(1-c)/2$ for any $c$) CAN is robust at every $c$ and *most* robust at the highest contention ($\rho$=1.20 at $c$=0.9). With *structurally capped* leverage (congestion) it is robust but cannot beat the $\rho$=2 ceiling by much at high $c$. With *weakening* leverage (stakes: contesting a jackpot at $c$=0.9 returns only 0.05 $v$) it degrades to $\rho$=4.05 at $c$=0.9—while still staying efficient (0.97) where the robust alternative (SOTO) delivers only 0.10. And with *absent* leverage—a rich-get-richer (Matthew) rule where the *richest* claimer wins, so a worst-off contester captures nothing and Prop. 1 fails by construction—CAN is exploited ($\rho$≈5.5), marking the boundary of the approach. The method works as

TABLE V: Generality: ($D$=0 efficiency, best-response $\rho$) per method, mean over $c$ (5 seeds). CAN is efficient ($\geq 0.94$) in every environment and the least exploitable efficient policy; SOTO's robustness costs catastrophic efficiency. $\rho_{\max}$ is the yield ceiling.

| | base ($\rho_{\max}$=6) | | congestion (2) | | stakes (6) | |
|---|---|---|---|---|---|---|
| Method | eff | $\rho$ | eff | $\rho$ | eff | $\rho$ |
| GGF | 1.00 | 6.00 | 1.00 | 2.01 | 0.98 | 5.75 |
| FEN | 0.81 | 4.20 | 0.96 | 2.05 | 0.70 | 3.27 |
| SOTO | 0.40 | 1.00 | 1.00 | 2.01 | 0.47 | 1.64 |
| **CAN** | **0.98** | **1.35** | **1.00** | **1.60** | **0.94** | **2.32** |

far as the leverage it is built on, and no further.

## VI. Discussion and Limitations

**What this shows.** The decentralization gap left open by prior work [5] is not fundamental: on the graded game a decentralized cross-attention policy trained against an adversarial league recovers most of the centralized oracle's incentive-compatibility ($\rho{\approx}1.2$–$1.5$) *and* its efficiency ($\approx 1.0$ when no free-rider is present), with no central allocator. The all-or-nothing futility result is the boundary case $c$=1. The effect is not automatic for every graded rule, though: it requires that contesting a free-rider actually return a positive share to the contester (Prop. 1), which the stakes and Matthew games show can be weak or absent.

**Where it is fragile.** (i) Zero-shot transfer to much larger teams degrades at high contention (Fig. 7), and a naive size curriculum does not fix it (it dilutes the per-size budget and leaves the $c$=0.9 gap unchanged); crowd-scale robustness at high waste is an open problem. (ii) The single-adversary scheme is seed-sensitive at low contention; league training is needed for consistency, at extra training cost. (iii) Robustness is bounded by leverage: on the stakes game it degrades at high contention (where contesting a jackpot returns almost nothing), and under a winner-take-all (Matthew) rule it disappears entirely—CAN is robust only as far as Prop. 1's leverage extends. (iv) Our games are controlled abstractions with a single shared cooperator policy; spatial/continuous environments are still open (coordinated defector coalitions, by contrast, we do test, and they are no harder than a lone free-rider). (v) The oracle remains an upper bound CAN approaches but does not reach.

**Takeaway.** Fair-MARL methods are usually reported on the fairness they achieve under full cooperation. We argue they should also be reported on the fairness they *retain* under self-interest—and that, contrary to the all-or-nothing intuition, a decentralized policy can retain much of it whenever the contention rule leaves a contester a real share to capture.

## VII. Conclusion

We gave a decentralized account of robust fairness in cooperative MARL. Graded contention makes contesting a free-rider strictly worthwhile for the worst-off (Prop. 1), but exploiting this requires inferring an unknown, variable number of free-riders and responding proportionally. CAN—a permutation-equivariant cross-attention policy over observed behaviour, trained against an adversarial league—does so, holding best-response exploitability low across the contention range (and most robust at the highest contention) while wasting essentially nothing when no free-rider is present, thereby approaching a centralized need-based oracle without a central allocator. We are explicit about scope rather than claiming blanket generality: across additional games CAN stays efficient and dominates the welfare-fair learners, but its robustness holds in proportion to the contest leverage—strong on a multi-server game, only partial where the leverage weakens (high-value steps at high contention), and absent under a winner-take-all rule. We hope this reframes decentralized robust fairness as a tractable, measurable target—bounded by a clear condition—rather than an impossibility.

## Code Availability

All code (the graded-contention game, CAN, the architecture and training-scheme ablations, and all experiments) is available in an anonymized repository for review at https://anonymous.4open.science/r/can-fair-marl; a de-anonymized release will follow upon publication.

## Use of AI-Assisted Technologies

The author used Claude (Anthropic), including Claude Code, to assist with the experimental code, manuscript review and revision, and formatting. All AI-assisted content was verified by the author, who is solely responsible for the methodology, results, and conclusions of this paper.

## Appendix A
## Environment Details

All environments share a skeleton: $N$=6 agents act for $T$=100 steps; at each step divisible resource is available; each agent chooses a self-interested act (Claim) or to Yield; accumulated utilities $\mathbf{u}$ determine fairness. They differ only in the *allocation rule* mapping the joint action to utility increments, and (for stakes/congestion) in the action space and one extra observation. Let $m$ be the number of claimers of a resource and $c \in [0,1]$ the contention waste.

**Base (single-resource graded contention).** One resource per step, binary action. Each claimer receives

$$g(m) = \begin{cases} 1 & m{=}1, \\ (1{-}c)/m & m \geq 2, \\ (\text{unit to worst-off } \arg\min_i u_i) & m{=}0, \end{cases}$$

so a sole claimer wins the unit, $m \geq 2$ claimers split a discounted $1{-}c$, and an unclaimed unit flows losslessly to the neediest agent. $c$=1 recovers the all-or-nothing model.

**Congestion (multi-server).** $M$=3 parallel unit resources ("servers") per step; action space $\{\text{Yield}, \text{claim } 1, \dots, \text{claim } M\}$ ($M$+1 actions). Each server is allocated independently by the base rule ($k$ claimers $\to 1$ if $k$=1, $(1{-}c)/k$ if $k \geq 2$); the $E$ empty servers route to the $E$ lowest-utility agents (one unit each, no waste). A lone free-rider can monopolise only one of $M$ servers, so the exploit ceiling is $\rho = N/M = 2$. Each token is augmented with the agent's per-server running claim-rates

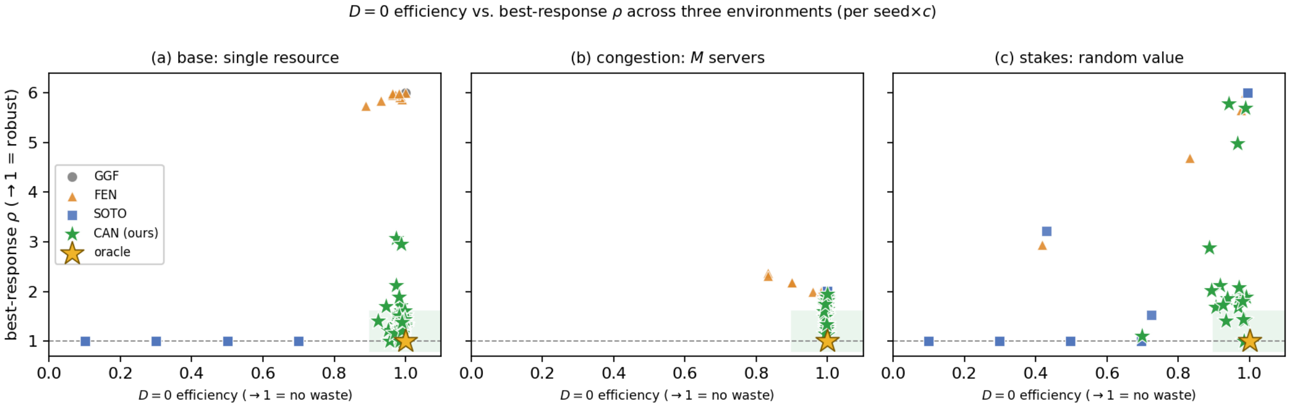


Fig. 8: $D$=0 efficiency vs. best-response $\rho$ on three structurally distinct leverage-preserving games (each marker is one seed×$c$). In every game CAN (green) stays on the efficient (right) side while remaining the least exploitable efficient policy; the only robust baseline, SOTO (blue), reaches low $\rho$ only by moving left into the wasteful region. GGF/FEN are efficient but exploitable (top-right). Shaded = the efficient-and-robust corner; gold star = centralized oracle.

(so the policy can infer *which* server is being grabbed): $5{+}M$ features, $(M{+}1)$-way head.

**Stakes (stochastic value).** One resource per step with random value $v_t$ drawn i.i.d. as $v_t$=3.25 w.p. 0.25 and $v_t$=0.25 otherwise (lumpy jackpots, $\mathbb{E}[v_t]$=1, so efficiency normalises as in base). The base rule scales by $v_t$: a sole claimer gets $v_t$, $m \geq 2$ claimers get $(1{-}c)v_t/m$ each, an unclaimed unit gives $v_t$ to the worst-off. The current $v_t$ is appended to each token (7 features), so the policy can contest selectively on high-value steps.

**Matthew (rich-get-richer; boundary).** One resource, binary action, but among claimers the *richest* wins: $m$=1 gives the sole claimer 1; $m \geq 2$ gives the richest claimer $(1{-}c)$ and the rest 0 (ties broken uniformly at random); $m$=0 routes to the worst-off. A worst-off cooperator that contests a richer free-rider receives 0, so Prop. 1 leverage vanishes; we use this environment only to mark the boundary of the method.

**Threat model and metric (all environments).** Each episode, $D$ agents are *defectors* with $D \sim \mathrm{Unif}\{0,\ldots,d_{\max}\}$ ($d_{\max}$=2 unless noted; the stress study uses $d_{\max}$=4); defectors always take the self-interested act and the $N{-}D$ cooperators share one policy. To audit a frozen cooperator policy we train a best-response defector (or a $D$-agent coalition sharing one policy) to maximise the defector group's utility, and report the bounded free-ride factor

$$\rho = \frac{N\sum_{i\in\mathcal{D}} u_i}{n_{\mathrm{def}}\sum_j u_j} \in [0,\ N/n_{\mathrm{def}}],$$

$\rho$=1 exactly fair, $\rho$=$N$ a lone defector taking everything. Cooperators maximise the per-step increment of $W_{\mathrm{coop}}$=$\mathrm{mean}_{i\in\mathcal{C}}(u_i){-}\mathrm{std}_{i\in\mathcal{C}}(u_i)$.

## Appendix B
## Implementation Details

CAN and all adversaries are shared single-head self-attention blocks (hidden width 64): $Q, K, V$ linear maps of the $N{\times}6$ token matrix, scaled-dot-product attention, a tanh layer on $[\,x_i \,\|\, \mathrm{ctx}_i\,]$, and a 2-logit head. All policies train with REINFORCE and Adam (lr $3\times10^{-3}$), $B$=512 parallel episodes, episodes of $T$=100 steps, with the entropy coefficient annealed from 0.05 to 0.003. Per-episode defector count is $D \sim \mathrm{Unif}\{0, 1, 2\}$. Vanilla/population co-training run 3000 updates; league training runs 6 generations of 1200 cooperator updates against the pool and 1000 updates per fresh best response. Every audit trains a best-response defector for 1500 updates against the frozen cooperator and reports $\rho$ from Eq. (2). The cooperator welfare is the masked $\mathrm{mean}{-}\mathrm{std}$ over cooperators; advantages are discounted ($\gamma$=0.99) per-step welfare increments with a batch-mean (or learned value) baseline. The architecture ablation reuses this league setup with the cross-attention block replaced by a mean-pool, deep-sets, or bi-GRU aggregator; the mixed-$N$ curriculum cycles the team size over $\{4, 6, 8, 12\}$ per update with the same total step budget. Unless noted, all numbers average 5 seeds and error bars are 95% bootstrap confidence intervals. League training is mildly sensitive to GPU non-determinism, which compounds over generations; the headline league/vanilla robustness (Table II, Fig. 4) therefore pools several independent gen-6 runs (15 and 10 seed-runs) so the CIs absorb run-to-run as well as seed variance.

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